125101 - Analytical chemistry 1 for engineers

Tutorial 9 – Potentiometry in Titration

T9-1. Given the following reaction

$$Fe(s) \mid Fe^{2+} \mid\mid Sn^{2+} \mid Sn(s)$$

a. Write the reaction of the anode;

b. Write the reaction of the cathode;

c. Write the net reaction.

Anode: in a galvanic cell, the anode is the electrode in which the oxidation occurs and the electrons leave away.

Cathode: the electrode that the reduction occurs and the electrons enter. Usually, we put the anode on the left hand, and the cathode on the right hand.

For Fe (s) | Fe^{2+} || Sn^{2+} | Sn (s)

a) Anode: Fe (s) $\rightleftharpoons$ Fe^{2+} + 2*e*

b) Cathode: Sn^{2+} + 2*e* $\rightleftharpoons$ Sn (s)

c) The net reaction: Fe (s) + Sn^{2+} $\rightleftharpoons$ Fe^{2+} + Sn (s)

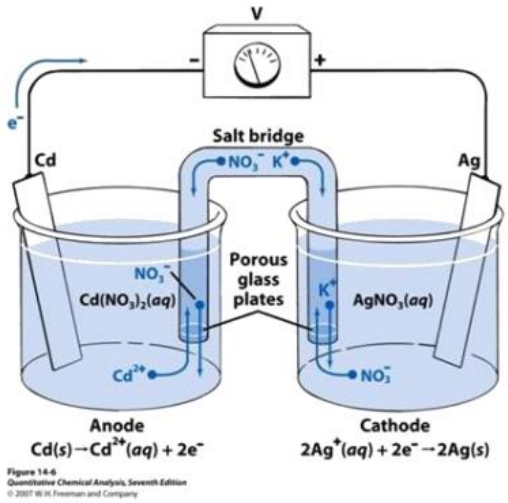
| stands for phase boundary

|| stands for salt bridge

For the electrodes, put them at the two sides of the notation.

If the compounds do not take part in the redox reaction, you can neglect them in the notation presentation.

T9-2. What is the cell potential E(cell) on the potentiometer measuring for the following set-up if the ion concentrations are all 1.0 mol/L (standard)?



$E^0(Cd^{2+}/Cd^0) = -0.402V$

$E^0(Ag^+/Ag^0) = 0.799V$

Line diagram for the above cell:

$Cd(s)|CdCl_2(aq)||AgNO_3(aq)|Ag(s)$

①Anode: Cd (s) ⇌ Cd^{2+} + 2*e* (pay attention to the direction of reaction, from valence of 0 to +2, oxidation occur)

E^o (Cd/Cd^{2+}) = -E^o (Cd^{2+}/Cd) = 0.402

②Cathode: $2Ag^+$ + 2*e* ⇌ 2Ag (s)

E^o (Ag^+/Ag) = 0.799

The net reaction①+②: Cd (s) + $2Ag^+$ ⇌ Cd^{2+} + 2Ag (s)

E^o(cell) = E^o (Cd/Cd^{2+}) + E^o (Ag^+/Ag) = 1.201 V

The potentiometer measures the difference in **standard <u>reduction</u> potential** (**reduction potential)** of the electrode attached to the positive terminal of the meter minus the **standard <u>reduction</u> potential** (**reduction potential)** of the electrode attached to the negative terminal of the meter. So, E(cell) = E^+ - E^- = 0.799 V - (-0.402 V) = 1.201 V

Note-1

Ordered redox potentials

Oxidizing agent	Reducing agent	E° (V)
$F_2(g) + 2e^- \rightleftharpoons 2F^-$		2.890
$O_3(g) + 2H^+ + 2e^- \rightleftharpoons O_2(g) + H_2O$		2.075
⋮		
$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O$		1.507
⋮		
$Ag^+ + e^- \rightleftharpoons Ag(s)$		0.799
⋮		
$Cu^{2+} + 2e^- \rightleftharpoons Cu(s)$		0.339
⋮		
$2H^+ + 2e^- \rightleftharpoons H_2(g)$		0.000
⋮		
$Cd^{2+} + 2e^- \rightleftharpoons Cd(s)$		−0.402
⋮		
$K^+ + e^- \rightleftharpoons K(s)$		−2.936
$Li^+ + e^- \rightleftharpoons Li(s)$		−3.040

Oxidizing power increases ⇑ (top to bottom)

Reducing power increases ⇓ (bottom to top)

When you reverse the direction of a reaction, reverse the sign of E°

If you reverse the direction of the reaction, reverse the sign of E^o. For example, E^o (Cd/Cd^{2+}) = - E^o (Cd^{2+}/Cd) = - (-0.402 V) = 0.402 V

Note-2

To multiply a factor in one half-reaction when it is necessary, do not multiply the factor with E^o.

For example,

①Anode: Cd (s) ⇌ Cd^{2+} + 2*e*

E^o (Cd/Cd^{2+}) = - E^o (Cd^{2+}/Cd) = 0.402

②Cathode: $2Ag^+$ + 2*e* ⇌ 2Ag (s)

E^o (Ag^+/Ag) = 0.799

The net reaction①+②: Cd (s) + $2Ag^+ \rightleftharpoons Cd^{2+}$ + 2Ag (s)

$E^o = E^o_{①} + 2 \times E^o_{②}$ = ??? why

$E_2 = E^0 - \frac{0.05916}{1} \frac{1}{[Ag^+]}$

$E_2' = E^0 - \frac{0.05916}{2} \frac{1}{[Ag^+]^2}$

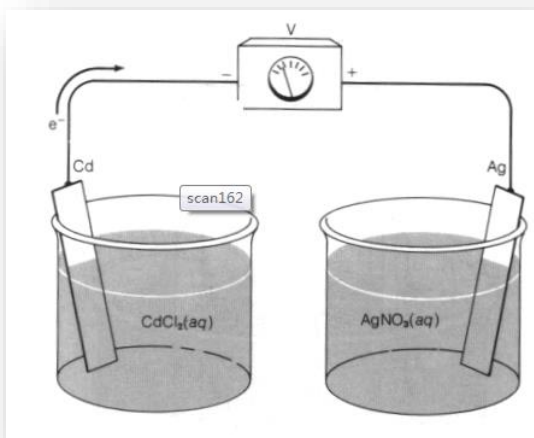
$E_2' = E_2$ because

$\frac{0.05916}{n} \log \frac{1}{[Ag^+]^n} = \cancel{n} \times \frac{0.05916}{\cancel{n}} \log \frac{1}{[Ag^+]} = 0.05916 \log \frac{1}{[Ag^+]}$

9T-3. Try to calculate the cell voltage in the following figure if the right half-cell contains 0.50 M $AgNO_3$ (aq) and the left half-cell contains 0.010 M $Cd(NO_3)_2$ (aq).

$$E^o(Ag^+|Ag) = 0.799 \text{ V}$$

$$E^o(Cd^{2+}|Cd) = -0.402 \text{ V}$$



(1) Calculation by two half-cells (two half-cells are **written as reduction forms**):

①Right electrode (Cathode):

$$2Ag^+ + 2e \rightleftharpoons 2Ag\ (s) \qquad E_+^0 = 0.799 \text{ V}$$

$$E_+ = E_+^0 - \frac{0.05916}{\cancel{2}} \log \frac{1}{[Ag^+]^{\cancel{2}}} = 0.799 - 0.05916 \log \frac{1}{0.50}$$

$$= 0.781 \text{ V}$$

②Left electrode (Anode):

$$Cd^{2+} + 2e \rightleftharpoons Cd\ (s) \qquad E_-^0 = -0.402\ V$$

$$E_- = E_-^0 - \frac{0.05916}{2} \log \frac{1}{[Cd^{2+}]} = -0.402 - \frac{0.05916}{2} \log \frac{1}{0.01}$$

$$= -0.461\ V$$

So, cell potential:

$$E = E_+ - E_- = 0.781 - (-0.461) = 1.242\ V$$

(2) If you write the ②Left electrode in its oxidation form:

$$Cd\ (s) \rightleftharpoons Cd^{2+} + 2e \qquad E' = -(-0.402\ V) = 0.402\ V$$

$$E'' = E' - \frac{0.05916}{2} \log [Cd^{2+}] = 0.402 - \frac{0.05916}{2} \log 0.01$$

$$= 0.461\ V$$

The net reaction $2Ag^+ + Cd(s) \rightleftharpoons 2Ag\ (s) + Cd^{2+}$, is ①+②

So, cell potential:

$$E = E_+ + E'' = 0.781 + 0.461 = 1.242\ V$$

Note, it is not necessary to multiply 2 for E_+ in the formula $E = E_+ + E''$.

(3) Or calculation using the net reaction:

$$Cd(s) + 2Ag^+ \rightleftharpoons 2Ag\ (s) + Cd^{2+}$$

$$E = \Delta E^0 - \frac{0.05916}{2} \log \frac{[Cd^{2+}]}{[Ag^+]^2}$$

$$= 0.799 - (-0.402) - \frac{0.05916}{2} \log \frac{0.01}{0.5^2} = 1.24\ V$$